

Colocalization-Confirmed Transcriptome-Wide Association Study of Ankylosing Spondylitis to Prioritize Tissue-Specific Drug Targets

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Introduction

Ankylosing spondylitis (AS) is a chronic inflammatory arthritis of the spine and sacroiliac joints affecting 0.1–0.5% of the population, with onset typically before age 30. GWAS have identified over 100 risk loci, but most fall in non-coding regions.

Because many AS-associated variants do not directly implicate a protein-coding gene, linking disease risk to *tissue-specific gene regulation* is an important step toward identifying biologically meaningful mechanisms and therapeutic targets.

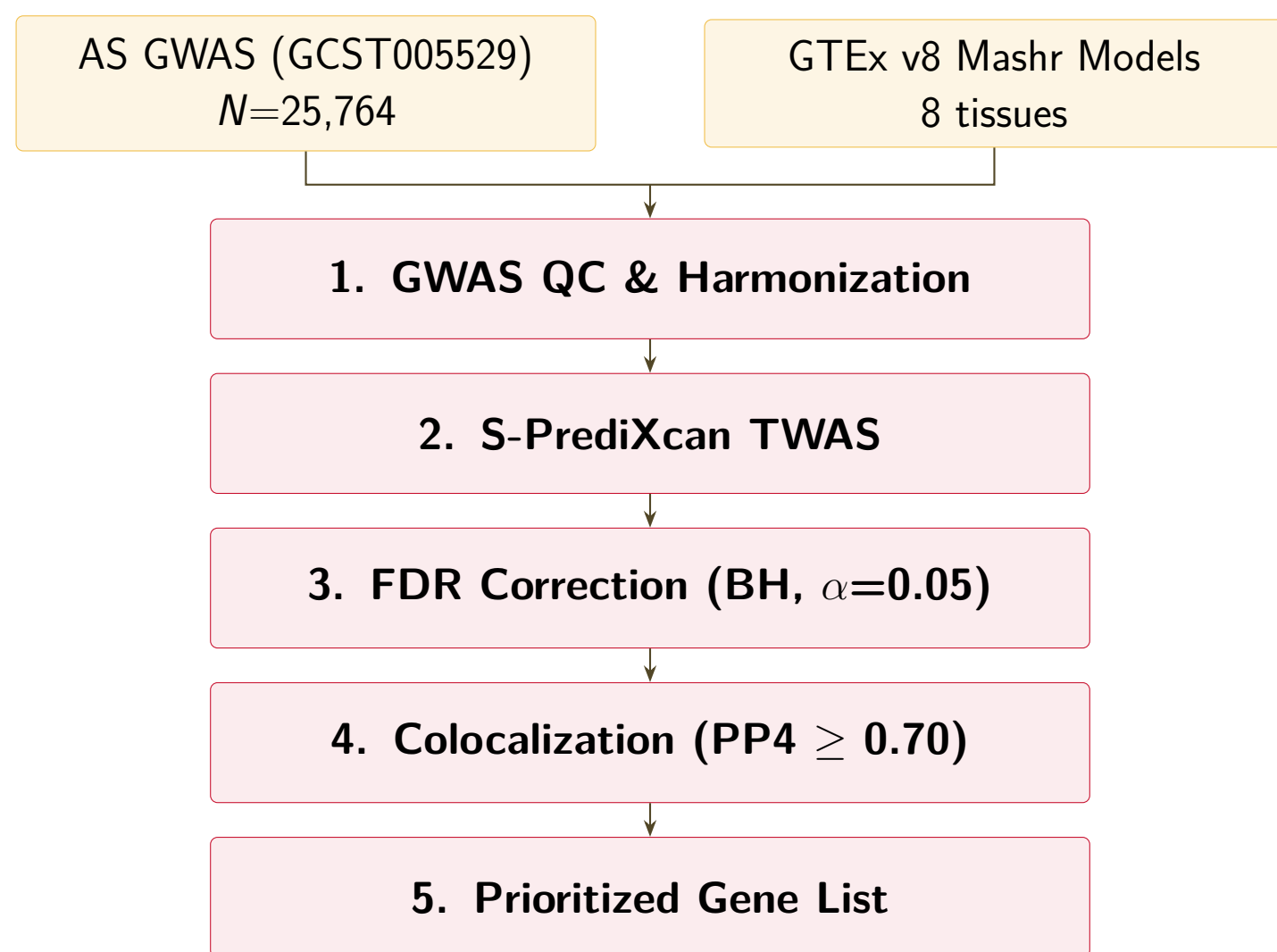
Transcriptome-wide association studies (TWAS) integrate GWAS with gene expression models to identify genes whose genetically regulated expression is associated with disease. However, TWAS hits can arise from *linkage disequilibrium (LD) contamination* rather than true shared causal variants.

Bayesian colocalization tests whether a GWAS signal and an eQTL signal share the same causal variant, helping distinguish high-confidence regulatory mechanisms from LD-driven associations.

Key Question

Which genes have genetically regulated expression that **causally colocalizes** with AS risk variants, and in which tissues?

Methods



GWAS QC: 123,941 SNPs → liftOver hg38 → 14,032 palindromic removed → 109,893 harmonized.
Tissues: Whole Blood, Spleen, Lung, Colon Sigmoid, Colon Transverse, Small Intestine, Muscle Skeletal, Adipose Subcutaneous.

Discussion

- ERAP1* (MHC-I antigen processing) is a known AS gene, validating our pipeline, and colocalizes in Whole Blood and Small Intestine.
- Novel candidates *GPR25*, *CCDC116*, *NPIP7* warrant experimental follow-up as tissue-specific drug targets.
- CARD9* and *FAS* implicate innate immune and apoptotic pathways beyond HLA-B27.
- Colocalization reduced 248 TWAS hits to 35 confirmed pairs, filtering LD-driven false positives.
- UBE2L3* (ubiquitin-proteasome pathway) may interact with *ERAP1* in antigen processing, suggesting a shared mechanistic axis.
- Whole Blood was the most frequently enriched tissue (11 of 35 pairs), consistent with AS being a systemic immune-mediated disease.

Predicted Protein Interaction Network

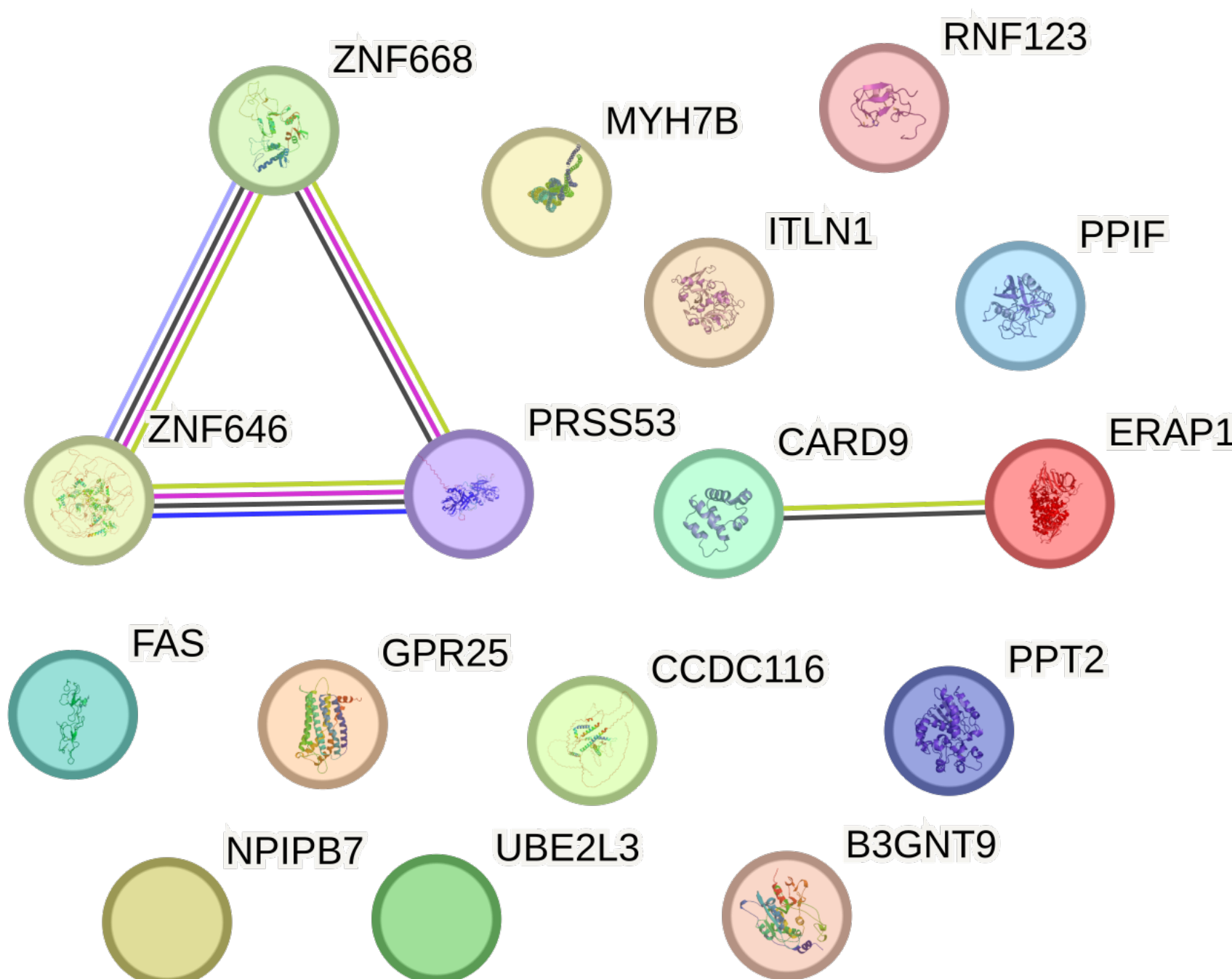


Figure. STRING-DB v12 subgraph of 16 prioritized proteins (score ≥ 0.400). Edges: *CARD9*–*ERAP1*, *ZNF646*–*ZNF668*, *PRSS53*–*ZNF646*/*ZNF668*.

TWAS Results

S-PrediXcan tested **2,313 gene-tissue pairs** across 8 GTEx v8 tissues. After Benjamini–Hochberg FDR correction ($\alpha = 0.05$), **248 significant associations** were identified, spanning genes on multiple chromosomes.

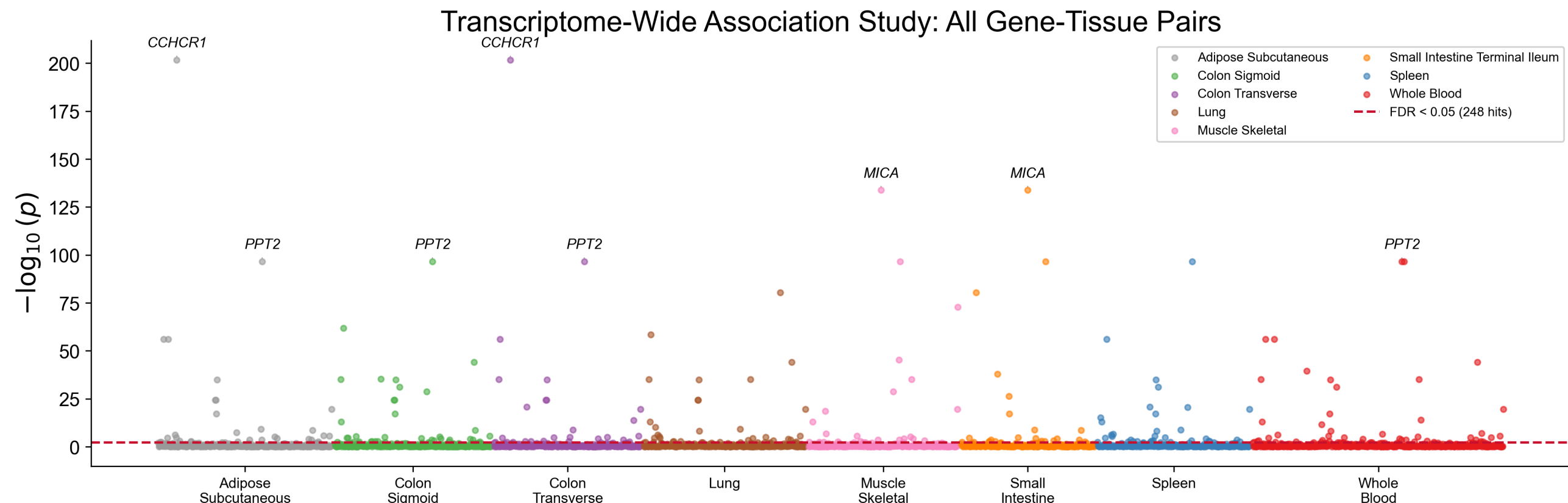


Figure. Manhattan-style plot of TWAS $-\log_{10}(p)$ across 8 tissues. Top gene symbols annotated. Dashed line: FDR < 0.05.

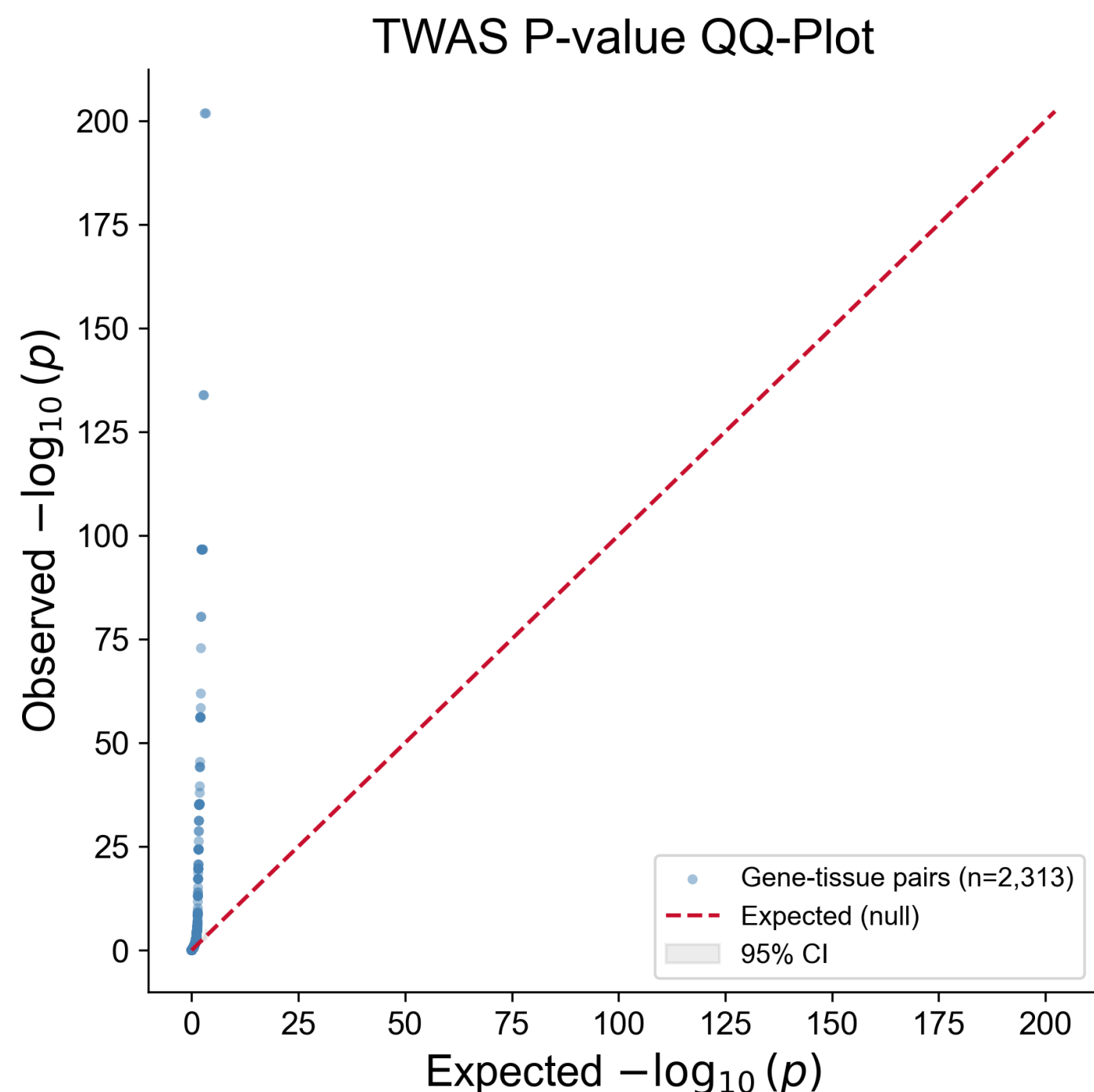


Figure. QQ-plot: departure from null consistent with polygenic architecture. Grey band: 95% CI.

Colocalization Results

Of the 248 TWAS hits, Bayesian colocalization (coloc.abf) identified **35 gene-tissue pairs (18 unique genes)** with $PP4 \geq 0.70$, confirming a shared causal variant between GWAS and eQTL signals.

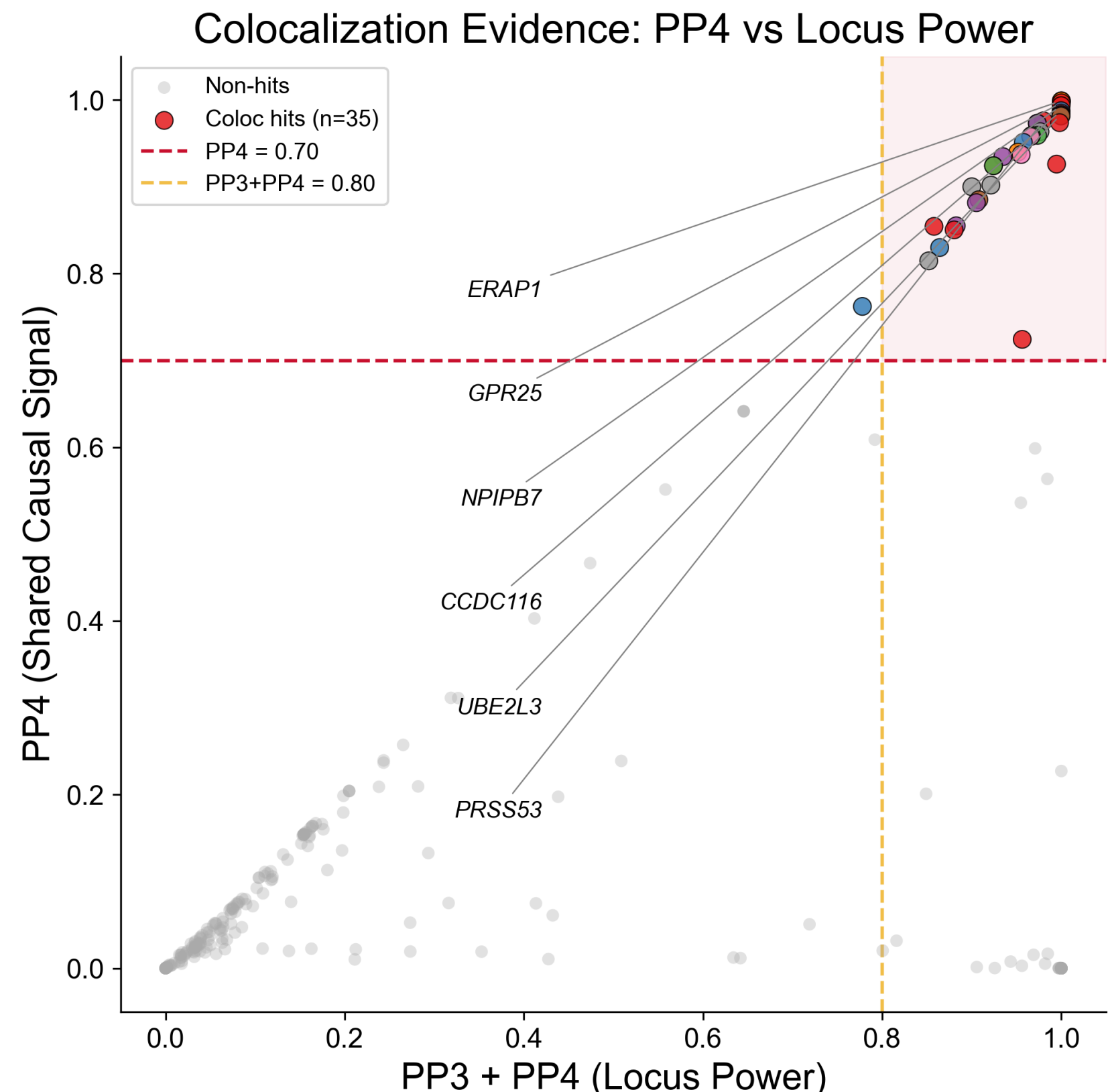


Figure. PP4 vs. locus power (PP3+PP4). Upper-right: colocalization-confirmed hits.

Conclusion

Integrating S-PrediXcan TWAS with Bayesian colocalization identified **18 unique genes** with shared causal variants at AS risk loci, prioritizing tissue-specific targets for therapeutic development.

This framework improves interpretability by narrowing a broad set of TWAS signals to a smaller group of higher-confidence gene-tissue pairs, providing a focused set of candidates for downstream fine-mapping and functional validation in AS.

Future Directions

- Residue-level binding-interface dissection of top STRING-predicted interactions via Seq2Bind [1]
- Fine-mapping (SuSiE) and validation in independent AS cohorts
- Drug repurposing analysis for top targets

Top Colocalized Genes

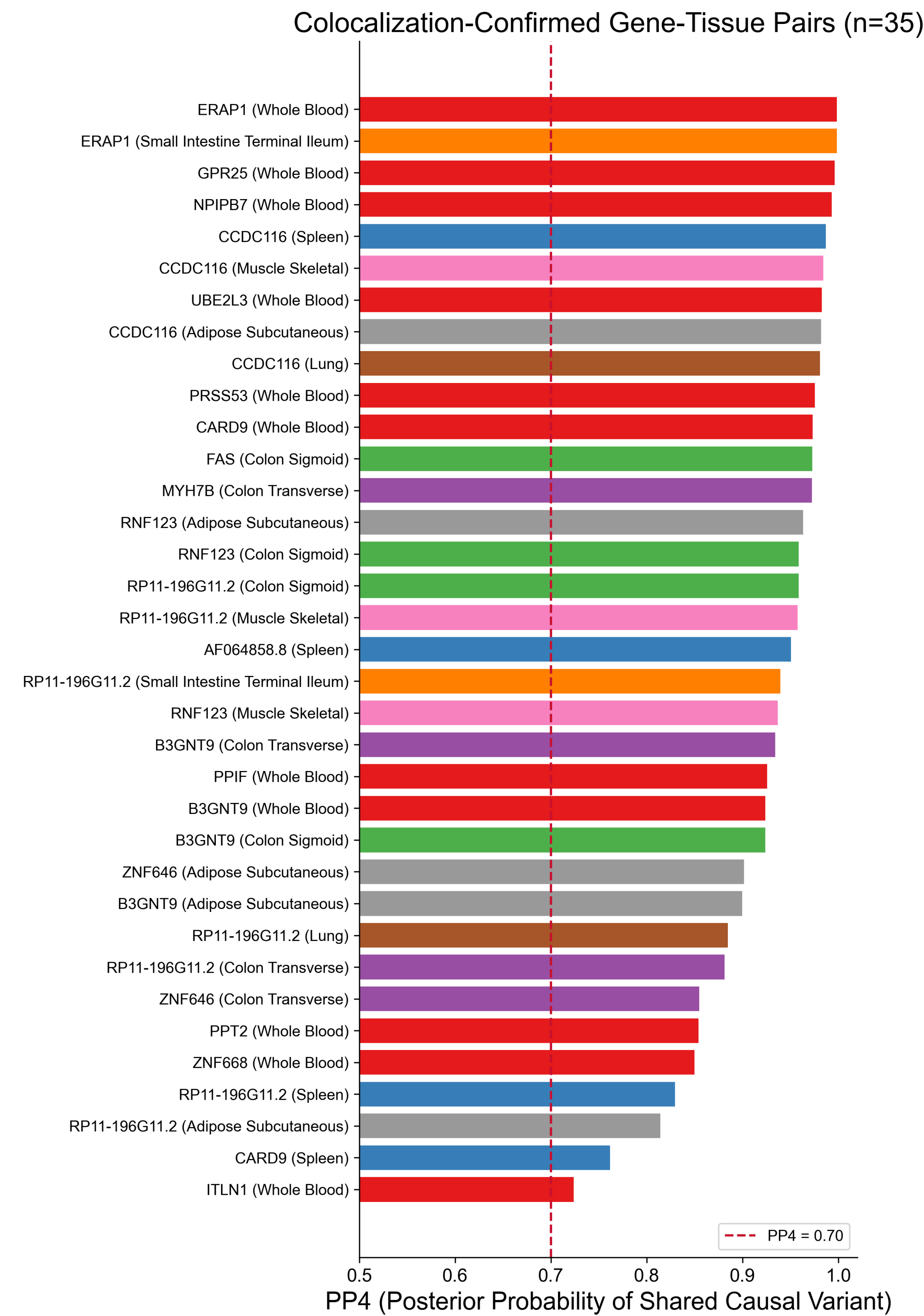


Figure. All 35 coloc-confirmed gene-tissue pairs ranked by PP4. Color indicates tissue. Dashed line: $PP4 = 0.70$.

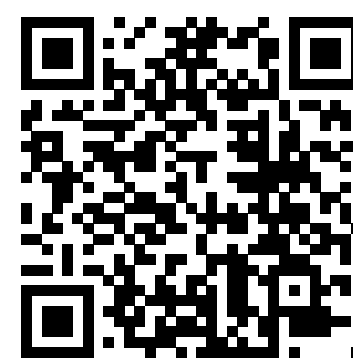
Gene	Tissue	PP4	z	FDR	p
<i>ERAP1</i>	Whole Blood	0.999	13.28	2.5e-38	
<i>GPR25</i>	Whole Blood	0.997	−7.03	6.0e-11	
<i>NPIP7</i>	Whole Blood	0.993	−5.16	5.9e-06	
<i>CCDC116</i>	Spleen	0.988	−5.16	5.9e-06	
<i>UBE2L3</i>	Whole Blood	0.983	5.32	2.5e-06	
<i>CARD9</i>	Whole Blood	0.974	4.94	1.7e-05	
<i>FAS</i>	Colon Sigmoid	0.973	4.42	1.9e-04	

Selected References

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Acknowledgements & Code

We thank **Dr. Ratul Chowdhury** for mentorship and project guidance. AS GWAS summary statistics were obtained from the NHGRI-EBI GWAS Catalog (GCST005529; Ellinghaus et al. 2016), with expression models from GTEx v8. Analyses used S-PrediXcan, coloc.abf, and STRING-DB v12 in Python and R.

github.com/yellepeddibk/as-twas-coloc